

TDVI: Inteligencia Artificial

Sesgos







tendemos a prestarle atención
a las cosas atípicas
(para nosotros)

Agenda

1. Bias
2. **Ejemplos**
3. Bias en AI
4. Métricas
5. Métodos

Apple Card Investigated After Gender Discrimination Complaints

A prominent software developer said on Twitter that the credit card was “sexist” against women applying for credit.

 Share full article



Jennifer Bailey, vice president of Apple Pay. Regulators are investigating Apple Card's algorithm, which is used to determine applicants' creditworthiness. Jim Wilson/The New York Times



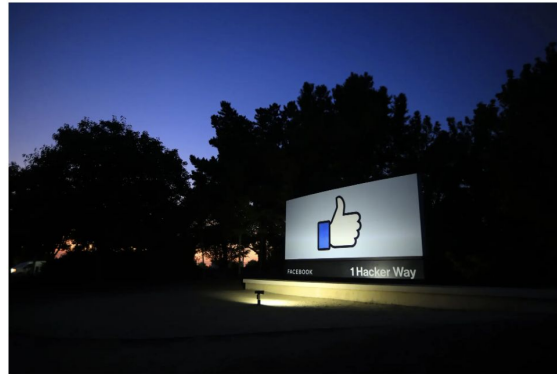
By **Neil Vigdor**

Nov. 10, 2019

Facebook Apologizes After A.I. Puts ‘Primates’ Label on Video of Black Men

Facebook called it “an unacceptable error.” The company has struggled with other issues related to race.

 [Share full article](#)  



Facebook apologized on Friday for mislabeling and said it was looking into its recommendation feature to “prevent this from happening again.” Jim Wilson/The New York Times



By [Ryan Mac](#)

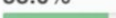
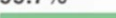
Published Sept. 3, 2021 Updated Oct. 4, 2021

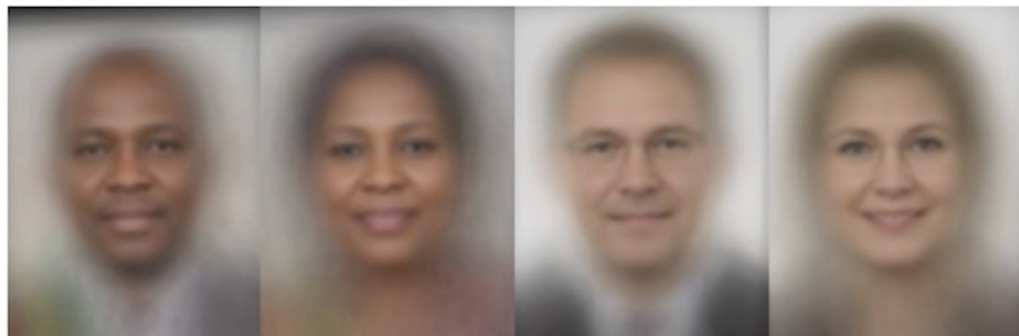
[Facebook users who recently watched videos from a British](#)

ON CAMPUS AND AROUND THE WORLD

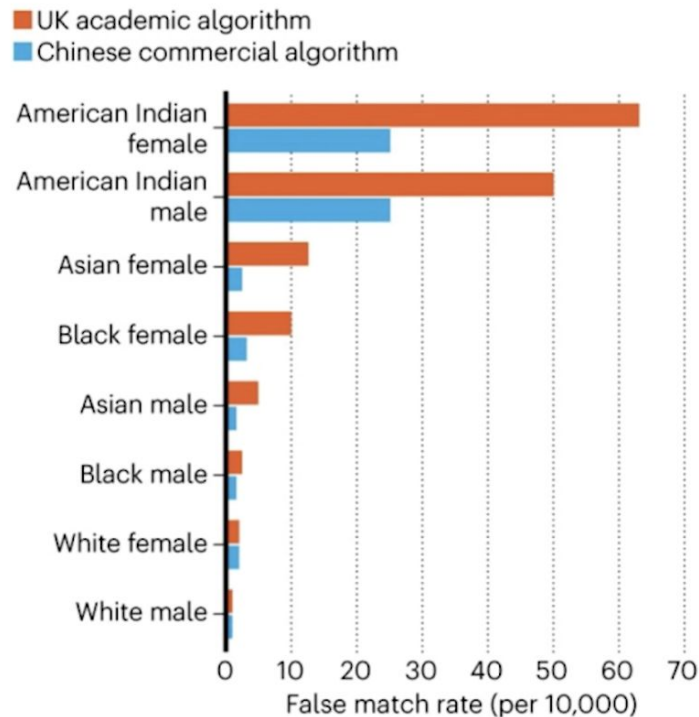
Bias in Facial Detection

Independent Study I

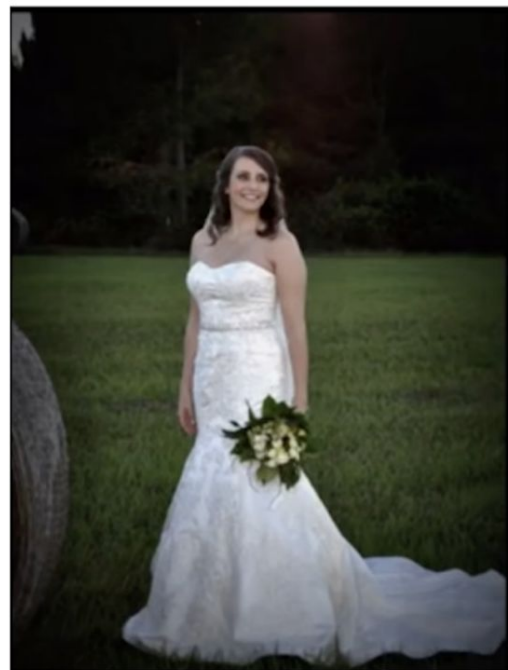
Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8% 
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8% 
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4% 



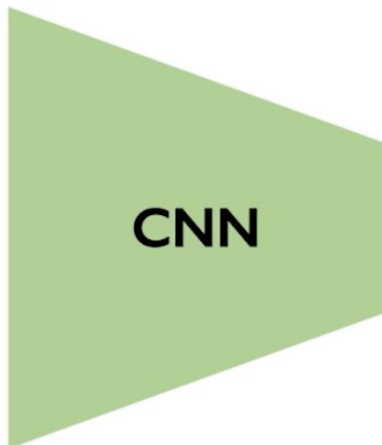
Independent Study II



Bias in Image Classification



Ground Truth: Bride



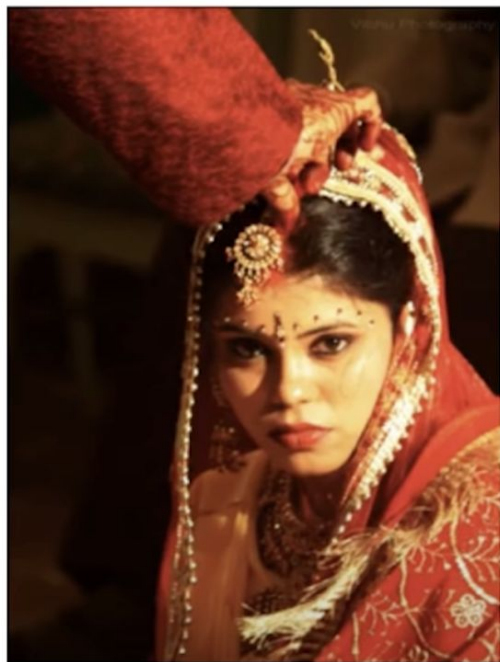
CNN for image
classification.



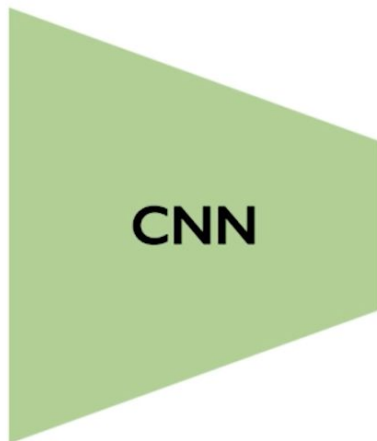
Predicted Classes

Bride
Dress
Ceremony
Woman
Wedding

Bias in Image Classification



Ground Truth: Bride



CNN

CNN for image
classification.



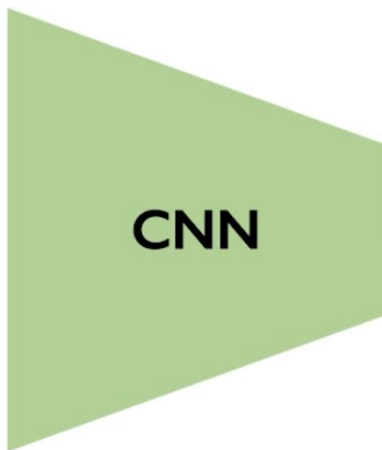
Predicted Classes

Clothing
Event
Costume
Red
Performance art

Bias in Object Recognition



Ground Truth: Spices



CNN

CNN for object
recognition.



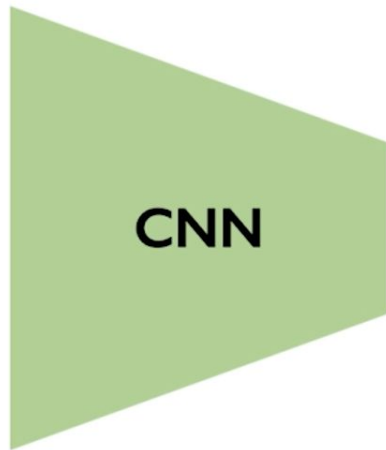
Predicted Objects

Seasoning
Spice
Spice rack
Ingredient

Bias in Object Recognition



Ground Truth: Spices



CNN for object
recognition.



Predicted Objects

Product
Yellow
Drink
Bottle

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Amazon scrapped 'sexist AI' tool

🕒 10 October 2018



The algorithm repeated bias towards men, reflected in the technology industry

An algorithm that was being tested as a recruitment tool by online giant Amazon was sexist and had to be scrapped, according to a Reuters report.

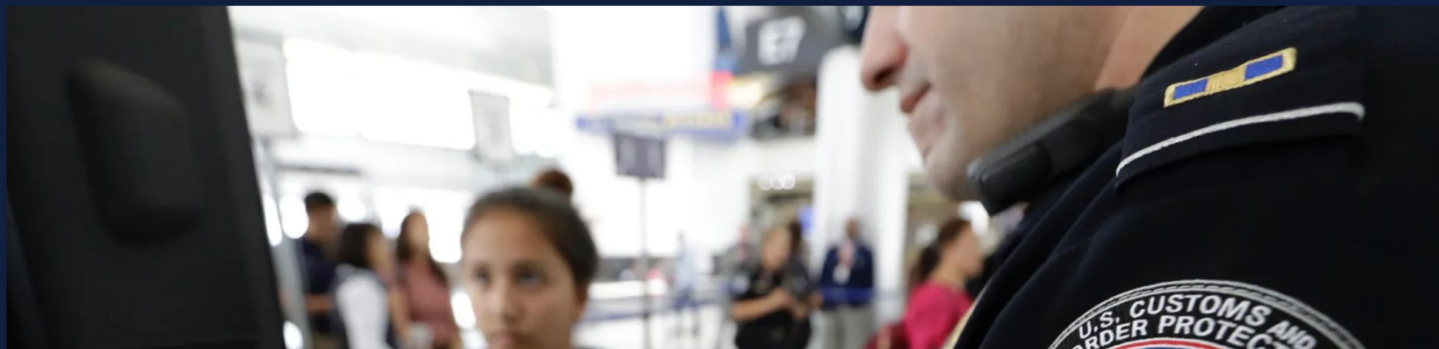
The artificial intelligence system was trained on data submitted by applicants over a 10-year period, much of which came from men, it claimed.

ARTIFICIAL INTELLIGENCE

A US government study confirms most face recognition systems are racist

By Karen Hao

December 20, 2019





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X

This article is more than 2 years old

Student proves Twitter algorithm 'bias' toward lighter, slimmer, younger faces

Company pays \$3,500 to Bogdan Kulynych who demonstrated flaw in image cropping software



A few of the generated faces that were used to test Twitter's image cropping algorithm.
Photograph: Bogdan Kulynych https://github.com/bogdan-kulynych/saliency_bias

Alex Hern

@alexhern

Tue 10 Aug 2021 13.00 BST



Most viewed



Violent protests in Dublin after woman and children injured in knife attack



Fiery Bentley crash that killed two at US-Canada border likely not terrorism, says FBI



Geert Wilders: the Dutch far-right figurehead sending a chill across Europe



OpenAI 'was working on advanced model so powerful it alarmed staff'



Revealed: King Charles secretly profiting from the assets of dead citizens

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica

May 23, 2016



“Scores like this — known as risk assessments — are increasingly common in courtrooms across the nation. They are used to inform decisions about who can be set free at every stage of the criminal justice system, from assigning bond amounts — as is the case in Fort Lauderdale — to even more fundamental decisions about defendants’ freedom. In Arizona, Colorado, Delaware, Kentucky, Louisiana, Oklahoma, Virginia, Washington and Wisconsin, the results of such assessments are given to judges during criminal sentencing.”

— Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner, ProPublica, May 23, 2016

(<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>)

Limited features

Diagnose heart disease

“recent research has shown that women who have heart disease experience different symptoms, causes and outcomes than men.”

“Women are often told their stress tests are normal or that they have “false positives.” Bairey Merz says doctors should pay attention to symptoms such as chest pain and shortness of breath rather than relying on a stress test score.”

**WHEN IT COMES
TO HEART DISEASE,
MEN AND WOMEN
ARE NOT EQUAL.**

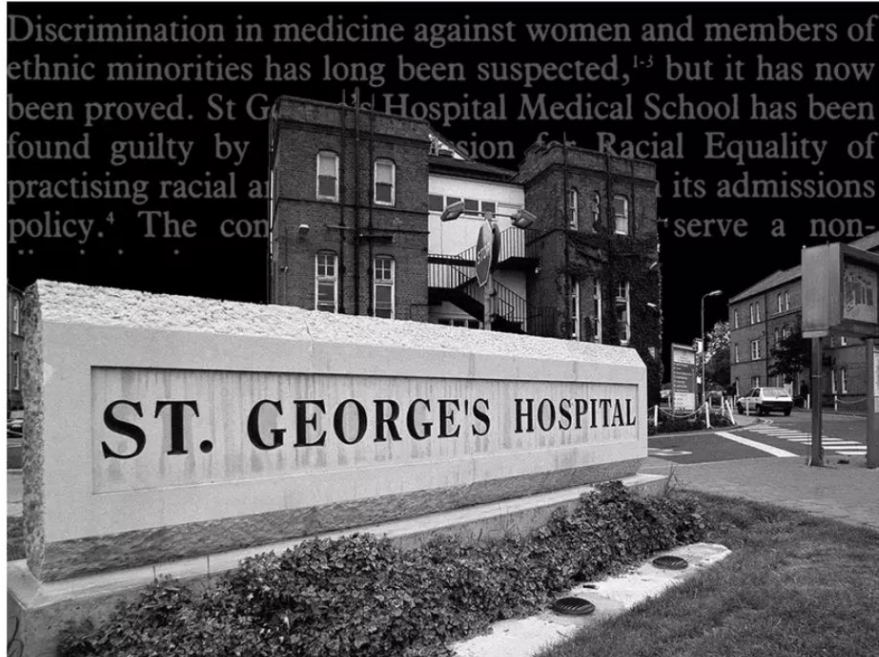
C. Noel Bairey Merz, MD, wants everyone to understand the differences between the sexes when it comes to treating heart disease.

[» Learn more](#)



Proxies

Make college admissions decisions



*“...certain rules in the system that weighed applicants on the basis of seemingly non-relevant factors, like **place of birth** and **name**.*

[...]simply having a non-European name could automatically take 15 points off an applicant's score. The commission also found that female applicants were docked three points, on average.”

- Oscar Schwartz, “Untold History of AI: Algorithmic Bias was Born in the 1980s”, April 15, 2019

Agenda

1. Bias
2. Ejemplos
3. **Bias en AI**
4. Métricas
5. Métodos

Sesgos en Todas las Etapas del Ciclo de Vida de la IA

1. **Datos:** desequilibrios con respecto a las etiquetas de clase, características, estructura de entrada.
2. **Modelo:** falta de incertidumbre unificada, interpretabilidad y métricas de rendimiento.
3. **Entrenamiento y despliegue:** bucles de retroalimentación que perpetúan los sesgos.
4. **Evaluación:** realizada de manera masiva, falta de análisis sistemático respecto a subgrupos de datos.
5. **Interpretación:** errores humanos y sesgos distorsionan el significado de los resultados.

¿Qué se necesita para confiar en una decisión tomada por una máquina?

(Además de que sea 99% precisa)

- ¿Es justa?
- ¿Es fácil de entender?
- ¿Alguien la manipuló?
- ¿Es responsable?

AI es una forma de discriminación estadística



Discrimination becomes objectionable when it places certain privileged groups at systematic advantage and certain unprivileged groups at systematic disadvantage

Illegal in certain contexts

Posibles fuentes de sesgos

Data-Driven

Selection Bias

Data selection does not reflect randomization
Ex: class imbalance

Reporting Bias

What is shared does not reflect real likelihood
Ex: news coverage

Sampling Bias

Particular data instances are more frequently sampled
Ex: hair, skin tone

Interpretation-Driven

Correlation Fallacy

Correlation $\neq$ Causation

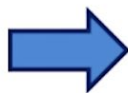
Overgeneralization

“General” conclusions drawn from limited test data

Automation Bias

AI-generated decisions are favored over human-generation decisions

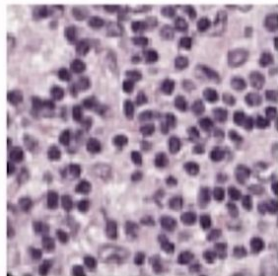
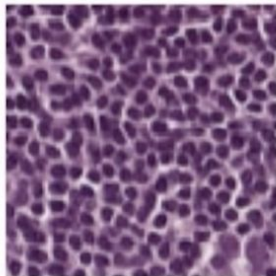
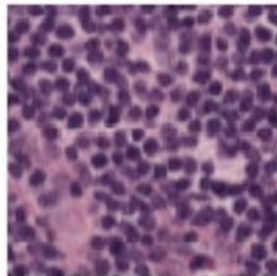
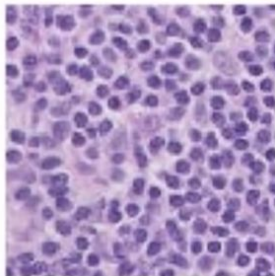
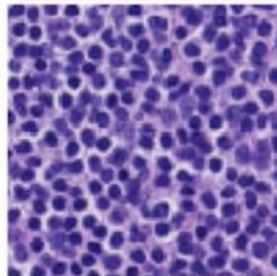
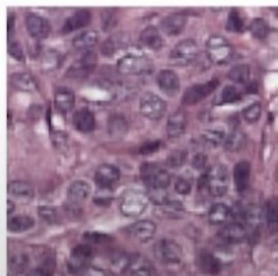
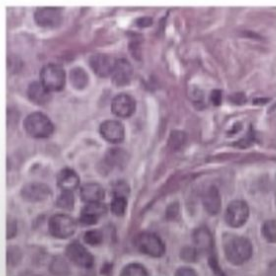
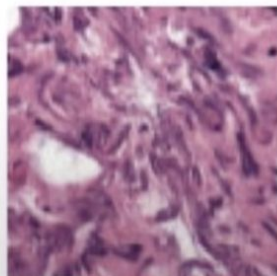
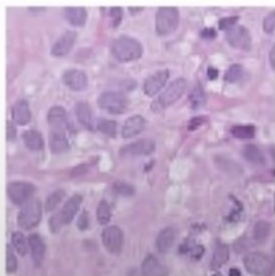
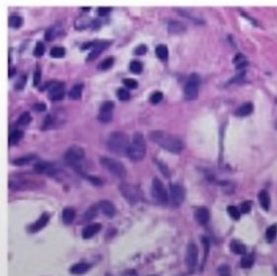
Expectation:
Cups in my dataset



Reality:
Cups from many angles



Distribution shift can result in neural network bias.

Train			Val (OOD)	Test (OOD)	
	$d = \text{Hospital 1}$	$d = \text{Hospital 2}$	$d = \text{Hospital 3}$	$d = \text{Hospital 4}$	$d = \text{Hospital 5}$
$y = \text{Normal}$					
$y = \text{Tumor}$					

Task: Disease classification from histopathology images

Distribution shift: Hospital source

Real world patterns of health inequality and discrimination



Unequal access and resource allocation



Discriminatory healthcare processes



Biased clinical decision making

Discriminatory data



Sampling biases and lack of representative datasets



Patterns of bias and discrimination baked into data distributions

Application injustices



Disregarding and deepening digital divides



Exacerbating global health inequality and rich-poor treatment gaps



Hazardous and discriminatory repurposing of biased AI systems

World → Data

Use ← Design

Biased AI design and deployment practices



Power imbalances in agenda setting and problem formulation

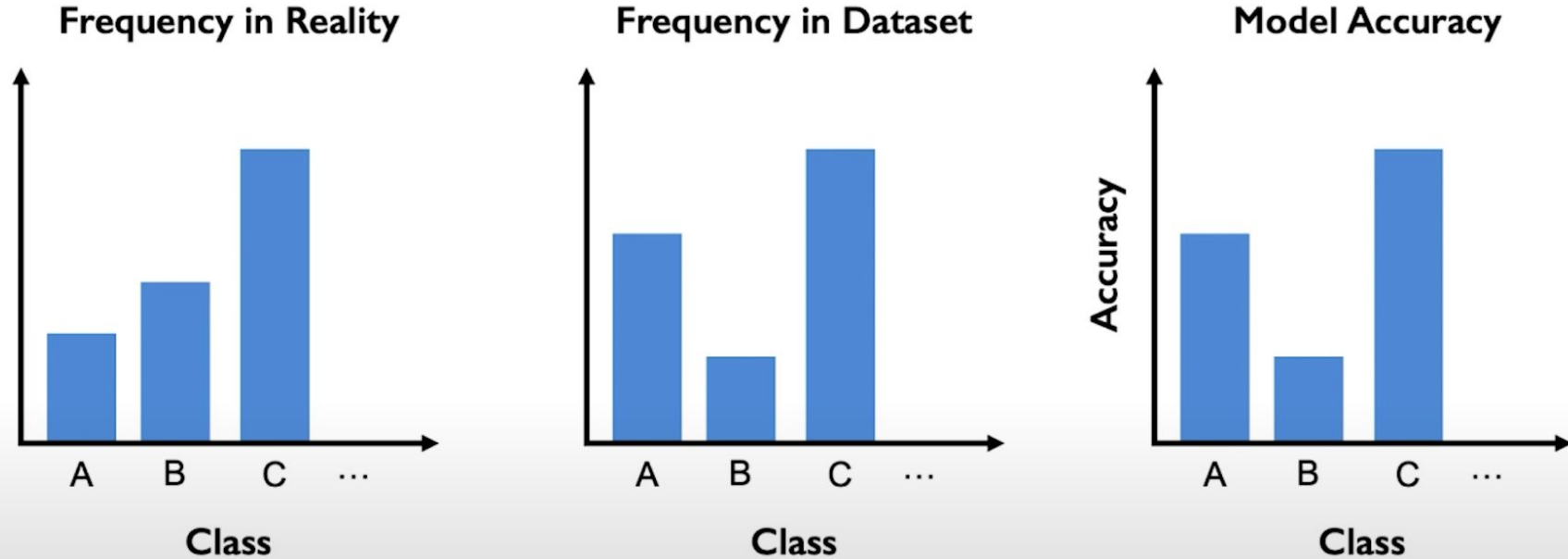


Biased and exclusionary design, model building and testing practices

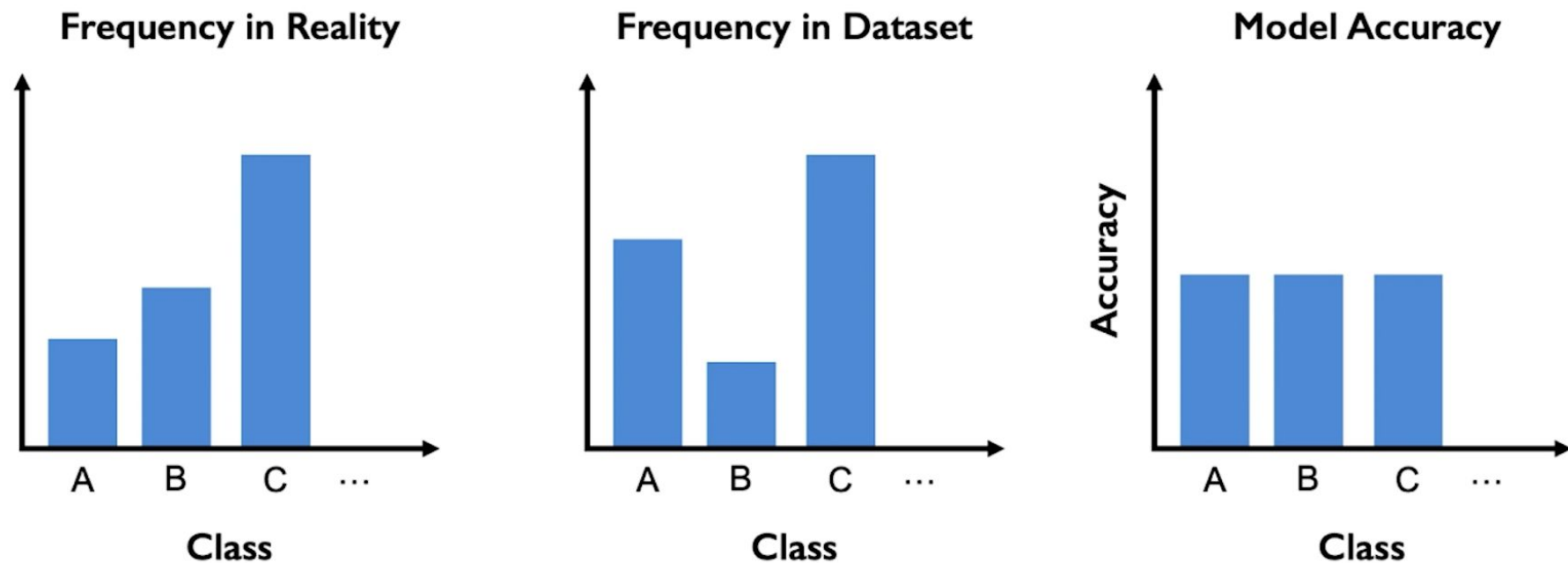


Biased deployment, explanation and system monitoring practices

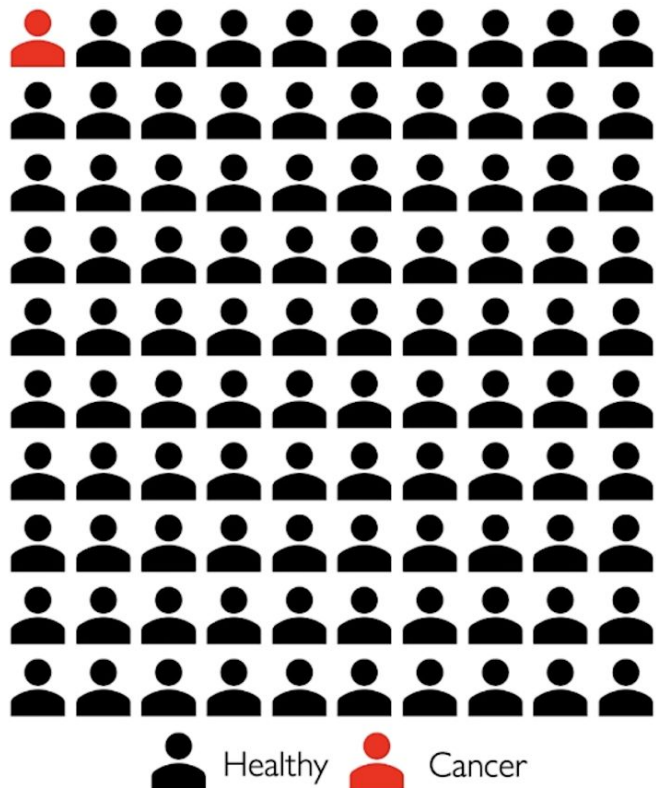
Biases due to Class Imbalance



Biases due to Class Imbalance



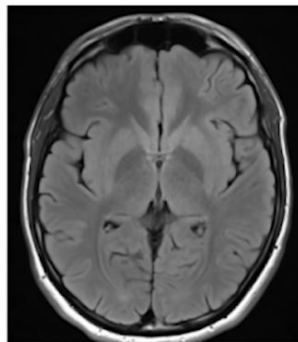
Case Study: The Danger of Class Imbalance



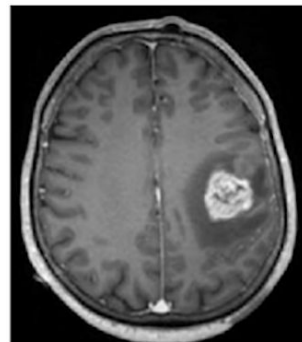
Case Study: Cancer Detection from Medical Images

- Glioblastoma (GBM): most aggressive and deadliest brain tumor
- GBM incidence in USA: **3.19 per 100,000** individuals!
- Task: train CNN to detect GBM from MRI scans of the brain

What if class incidence in dataset reflected real-world incidence?



99.997%



0.003%

Agenda

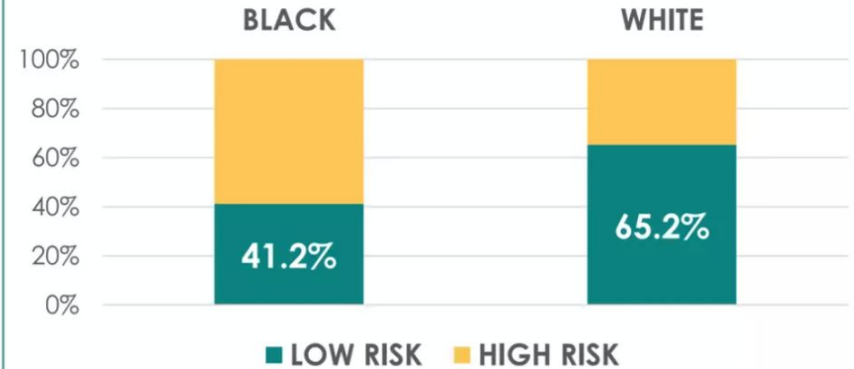
1. Bias
2. Ejemplos
3. Bias en AI
- 4. Métricas**
5. Métodos

Fairness Metrics

Statistical Parity Difference

Difference of the rate of favorable outcomes between the unprivileged and privileged groups

Statistical Parity Difference = -24.0%

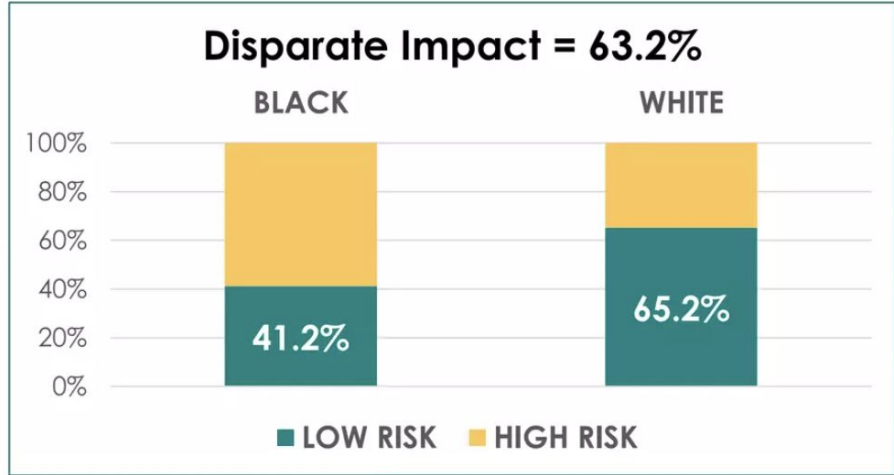


ALL DEFENDANTS			BLACK DEFENDANTS			WHITE DEFENDANTS		
	Low	High		Low	High		Low	High
Survived	2681	1282	Survived	990	805	Survived	1139	349
Recidivated	1216	2035	Recidivated	532	1369	Recidivated	461	505

Fairness Metrics

Disparate Impact

Ratio of the rate of favorable outcomes between the unprivileged and privileged groups

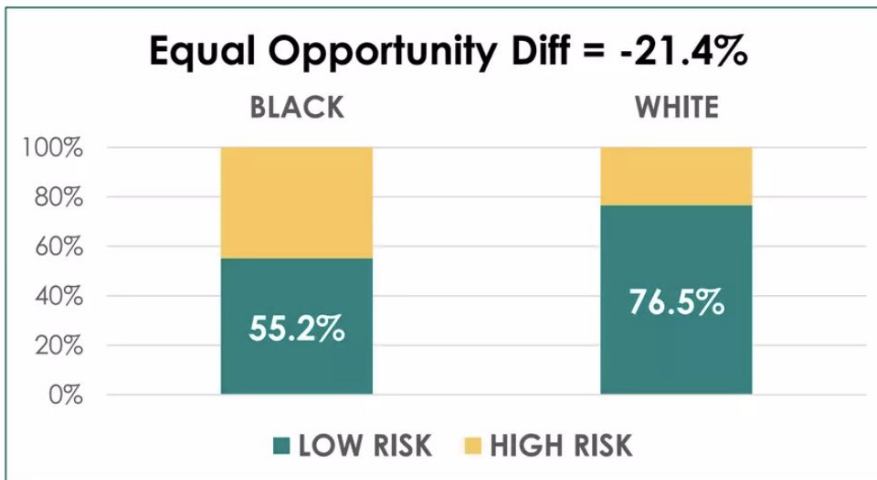


ALL DEFENDANTS			BLACK DEFENDANTS			WHITE DEFENDANTS		
	Low	High		Low	High		Low	High
Survived	2681	1282	Survived	990	805	Survived	1139	349
Recidivated	1216	2035	Recidivated	532	1369	Recidivated	461	505

Fairness Metrics

Equal Opportunity Difference

Difference in true positive rates between the unprivileged and privileged groups

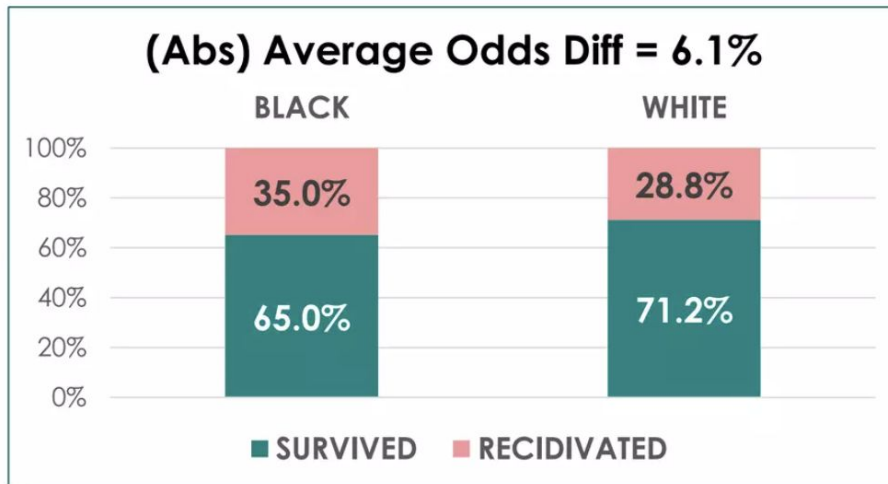


ALL DEFENDANTS			BLACK DEFENDANTS			WHITE DEFENDANTS		
	Low	High		Low	High		Low	High
Survived	2681	1282	Survived	990	805	Survived	1139	349
Recidivated	1216	2035	Recidivated	532	1369	Recidivated	461	505

Fairness Metrics

Average Odds Difference

Average of difference in false positive rate and true positive rates of favorable outcomes between the unprivileged and privileged groups



ALL DEFENDANTS			BLACK DEFENDANTS			WHITE DEFENDANTS		
	Low	High		Low	High		Low	High
Survived	2681	1282	Survived	990	805	Survived	1139	349
Recidivated	1216	2035	Recidivated	532	1369	Recidivated	461	505

<https://aif360.res.ibm.com/>

<https://github.com/Trusted-AI/AIF360>

Agenda

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Mitigate Bias

SEBASTIEN
@RTPAr

Pre-processing

Mitigate bias in training data

- Disparate Impact Remover
- Learning Fair Representations
- Optimized Preprocessing
- Reweighing

In-processing

Mitigate bias in classifiers

- Adversarial Debiasing
- ART Classifier
- Prejudice Remover

Post-processing

Mitigate bias in predictions

- Calibrated Equality of Odds
- Equality of Odds
- Reject Option Classification

Métodos para mitigar bias

Predecimos la tarea + la variable sensible

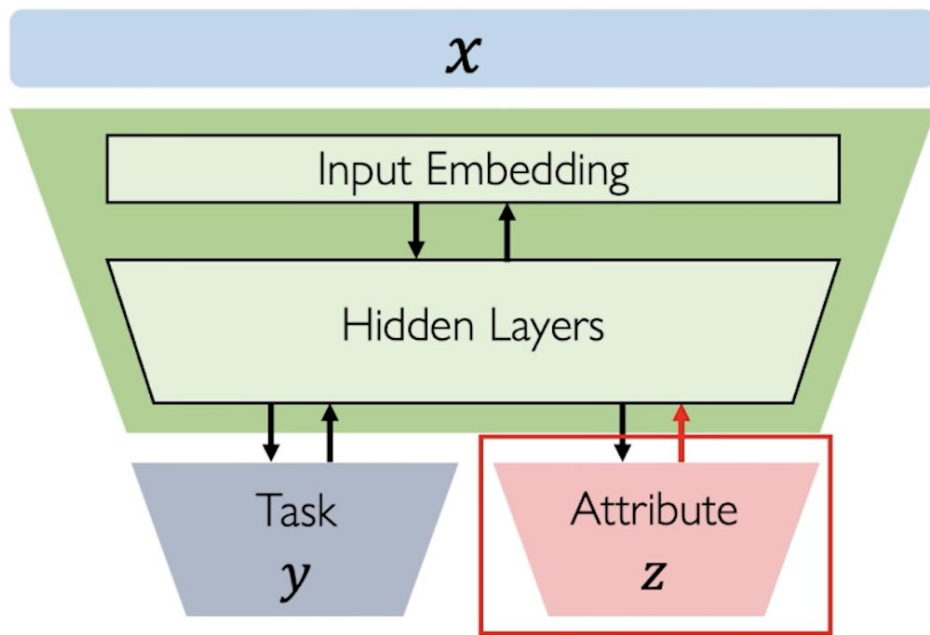
La variable sensible se niega durante backpropagation para eliminar el impacto.

Task: language model to complete analogies

He is to **she**, as **doctor** is to ?

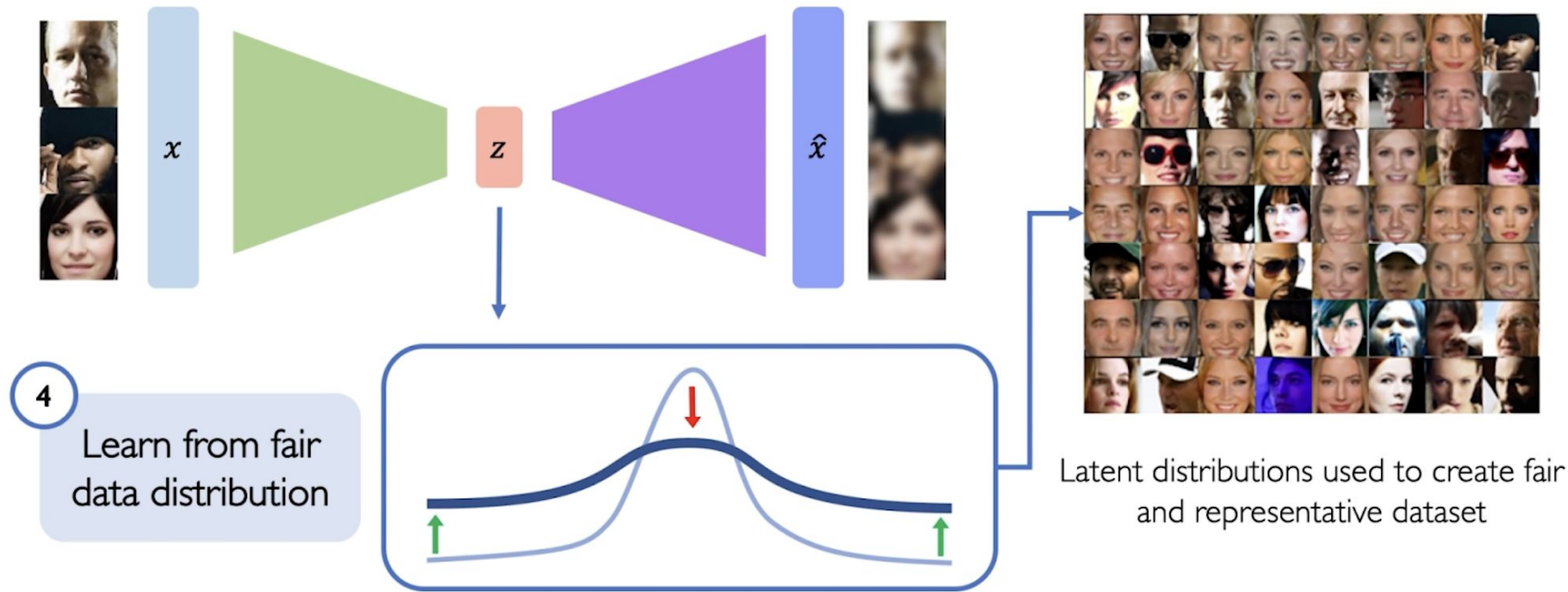
biased		debiased	
neighbor	similarity	neighbor	similarity
nurse	1.0121	nurse	0.7056
nanny	0.9035	obstetrician	0.6861
fiancée	0.8700	pediatrician	0.6447
maid	0.8674	dentist	0.6367
fiancé	0.8617	surgeon	0.6303
mother	0.8612	physician	0.6254
fiance	0.8611	cardiologist	0.6088
dentist	0.8569	pharmacist	0.6081
woman	0.8564	hospital	0.5969

Sensitive attribute: Gender



Jointly predict output label y and sensitive attribute z to remove from decision

Mitigating Bias through Learned Latent Structure



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6. **Fin**